

Acids and bases

What are the Brønsted-Lowry and Lewis definitions of the terms *acid* and *base*? [5]

Brønsted-Lowry

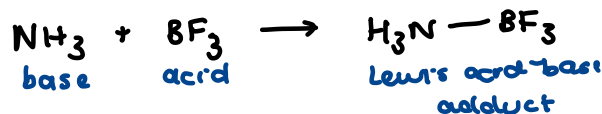
Acid = proton donor

Base = proton acceptor

Lewis

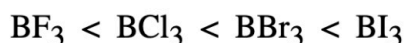
Acid = electron pair acceptor

Base = electron pair donor



Account for the following observations:

The Lewis acidity of the boron trihalides is as follows:



but the trend in acidity for the silicon tetrahalides is

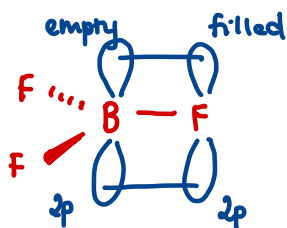


electronegativity increases
more electron accepting

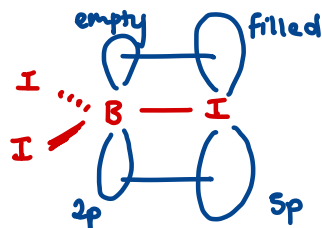


[5]

Acidity of boron trihalides depends on π -backbonding ($p\pi-p\pi$ bonding)



most backbonding



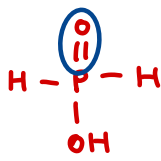
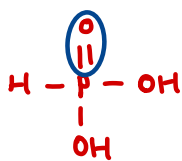
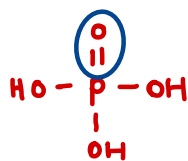
least backbonding

Poorer energy match ($2p/5p$)
Worse backbonding $\therefore \text{BI}_3$
more Lewis acidic

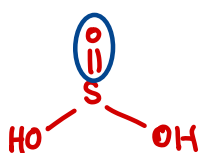
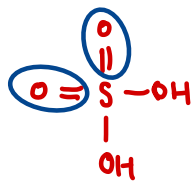
H_3PO_4 , H_3PO_3 and H_3PO_2 have similar first acid dissociation constants ($\text{pK}_a \sim 2$)

whereas H_2SO_3 and H_2SO_4 have first acid dissociation constants (pK_a) of 1.9 and -2.4 respectively.

[6]



All have same number of oxo groups $\rightarrow$ same delocalisation in conjugate base
(Estimate $\text{pK}_a \approx 3$)

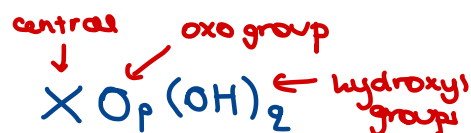


H_2SO_4 has 2 oxo groups vs 1 in H_2SO_3
More delocalisation in conjugate base
(Estimate $\text{pK}_a \approx -2$ and $\text{pK}_a \approx 3$)

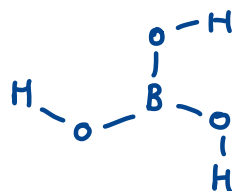
Pauling's rules : estimate pK_a of oxoacids

$$\text{pK}_a \approx 8 - 5p$$

Each successive deprotonation increases pK_a by 5 units



3) Boric acid, H_3BO_3 , is a weak monobasic acid with a pK_a value of 9.2. [3]

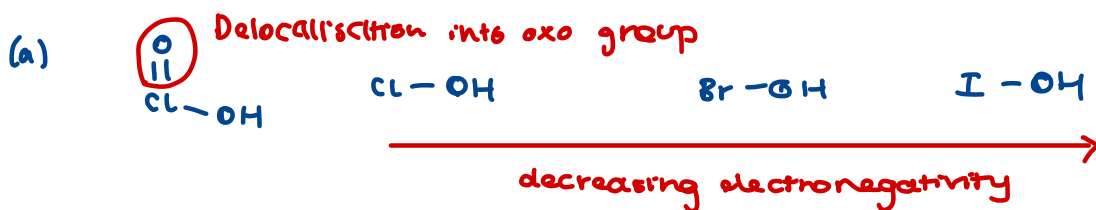
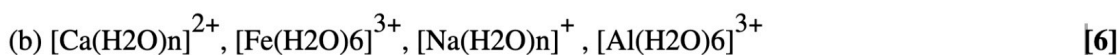
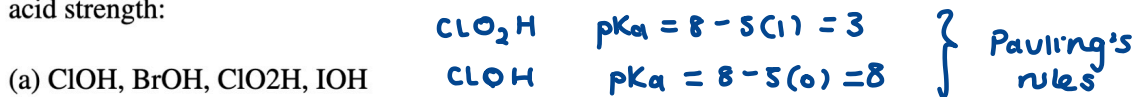


Lewis acid

Accepts electron pair into empty p-orbital

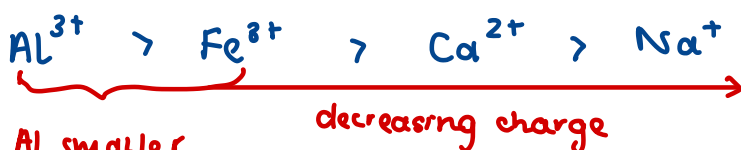
$$\text{pK}_a = 8 - 5 \times 0 = 8$$

Giving reasons for your choice, arrange the following two series in order of increasing acid strength:



(b) Charge on metal polarises O-H bond and repels proton as it leaves

Greater charge density on metal $\rightarrow$ more acidic complex

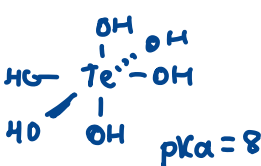
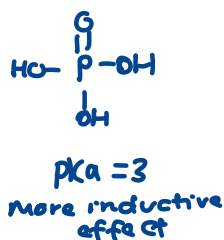
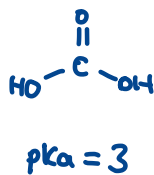
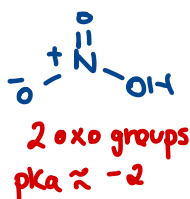


$$\text{charge density} = \frac{\text{charge}}{\text{volume}}$$

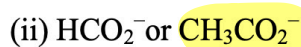
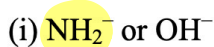
All smaller ionic radius
 $\therefore$ higher charge density

Rationalize the first pK_a values of the following species:

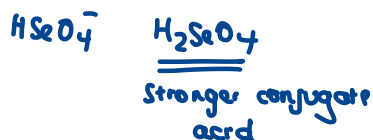
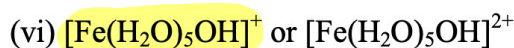
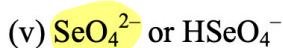
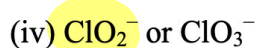
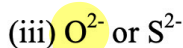
Pauling's acidity rule



(b) For **each** of the following pairs of ions, identify the stronger Brønsted base, giving reasons for your choice:



Weaker conjugate acid
 $\therefore$ stronger base



[6]